ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

SEMESTER FINAL EXAMINATION **WINTER SEMESTER, 2022-2023**
DURATION: 3 HOURS **FULL MARKS: 150**

CSE 4303: Data Structures

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 6 (six)** questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

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1. a) Professor Bunyan thinks he has discovered a remarkable property of Binary Search Trees (BST). Suppose that the search for key k in a BST ends up in a leaf. Consider 3 sets: A, the keys to the left of the search path; B, the keys on the search path; and C, the keys to the right of the search path. Professor Bunyan claims that any three keys $a \in A$, $b \in B$, and $c \in C$ must satisfy $a \leq b \leq c$. Provide a counterexample disproving the professor's claim. 6 (CO1) (PO1)

   b) Suppose a Priority Queue is implemented using a balanced BST. Discuss how the following requirements can be satisfied: 3×3 (CO1) (PO1)
      i. Extract the element with the highest priority
      ii. Insert an element with an arbitrary priority
      iii. Modify the priority of an existing element

   c) Prove that, the process of building a Max-heap from an arbitrary set of numbers can be done in linear time. 10 (CO3) (PO2)

2. a) Professor Marley hypothesizes that he can obtain substantial performance gains by modifying the separate chaining scheme to keep each list in sorted order. How does the professor's modification affect the running time for successful searches, unsuccessful searches, insertions, and deletions? 5 (CO3) (PO2)

   b) Supposed a hash table is maintained by adopting an open addressing technique as the collision resolution strategy. Identify the potential problem(s) of deleting an element from the table and propose solution(s) to improve the situation. 5 (CO3) (PO2)

   c) Consider inserting the keys $\{10, 22, 31, 4, 15, 28, 17, 88, 59\}$ into a hash table of length $m = 11$, where the hash function is defined as $h(x) = x \; \% \; m$. Illustrate the steps of inserting the keys using linear probing, quadratic probing, and double hashing. Utilize a secondary hash function $h_2(x) = 1 + (x \; \% \; (m-1))$ where necessary. 5×3 (CO2) (PO1)

3. a) Suppose the Disjoint Set data structure is applied to determine the connected components of an undirected graph $G = (V, E)$, where $V = \{a, b, c, d, e, f, g, h, i, j, k\}$. Showing detailed steps, list the vertices of each connected component by processing the edges of E in the order $(d, i), (f, k), (g, i), (b, g), (a, h), (i, j), (d, k), (b, j), (d, f), (g, j)$, and (a, e). 12 (CO2) (PO1)

   b) To implement *union-by-height*(i, j), the heights of the trees containing i and j need to be compared. When the tree is modified by a union operation, how much time will it take to modify the heights of all the nodes that need to be modified? Assume the heights are stored with each node. 6 (CO3) (PO2)

4. a) Suppose N integers are stored in an array. To find the maximum XOR value between two elements of this array, one obvious way is to compute the XOR of every pair of elements and identify the maximum, resulting in an algorithm of $O(N^2)$ time complexity. Propose an improved approach using the Trie data structure and simulate considering the array as $\{9, 3, 10, 5, 25\}$. 10+5 (CO3) (PO2)

b) What is the benefit of using the Segment Tree data structure? Explain the concept of lazy propagation in the context of a segment tree. 10 (CO1) (PO1)

5. a) How can topological sort be used to detect cycles in a directed graph? 5 (CO1) (PO1)

b) Suppose we wish to search a linked list of length n, where each element contains a key k representing a long character string. To find a query string from this list, one obvious approach is to check every node by comparing the characters, requiring $O(kn)$ time in the worst case. Propose a better idea without using any additional data structure. 5 (CO3) (PO2)

c) Define Data Structures. Mention one use case for each of the following data structures: Stack, Deque, Linked List, Priority Queue, AVL Tree, Hash Table, Trie, and Graph. $2 + 8$ (CO1) (PO1)

d) Assume some numbers are stored in a Linked List (LL). Mention the worst-case running time for the operations mentioned in Table 1. 24×0.5 (CO1) (PO1)

	unsorted, singly LL	sorted, singly LL	unsorted, doubly LL	sorted, doubly LL
Search (key)				
Insert (x)				
Delete (x)				
Successor (x)				
Predecessor (x)				
Minimum ()				
Maximum ()				

Table 1: Incomplete worst-case run time table for Question 5.d)

6. a) Showing detailed steps, find the distance (d) and predecessor (π) values for each vertex that result from running a Breadth-first search (BFS) on the directed graph given in Figure 1. Use vertex q as the source. The BFS procedure will consider the vertices in alphabetical order and assume that each adjacency list is ordered alphabetically. 16 (CO2) (PO1)

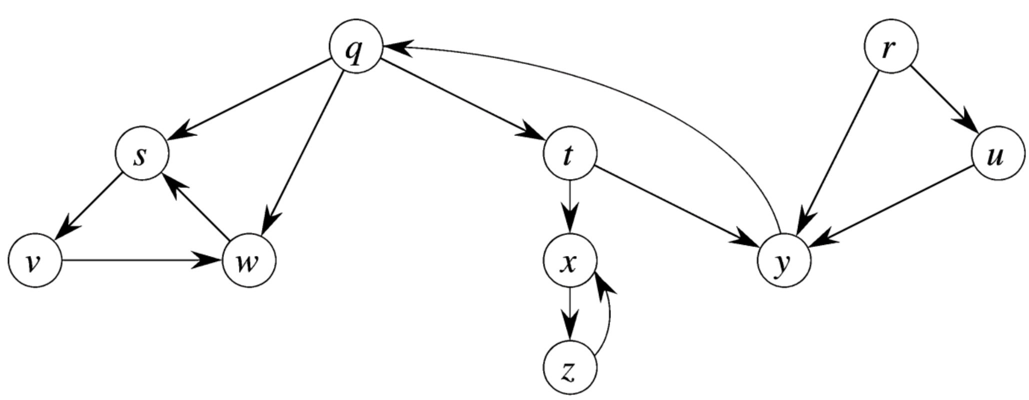

Figure 1: A directed graph for Question 6.a)

b) Suppose Depth-first Search is applied on a graph G, and the discovery and finishing time for every vertex are stored using the attributes d & f, respectively. Justify that an edge (u, v) is: 3×3 (CO1) (PO1)

i. A tree edge or forward edge if and only if $u.d < v.d < v.f < u.f$

ii. A back edge if and only if $v.d \leq u.d < u.f \leq v.f$

iii. A cross edge if and only if $v.d < v.f < u.d < u.f$

B.Sc. Engg. CSE 3rd Semester **9 December 2022**

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

SEMESTER FINAL EXAMINATION **WINTER SEMESTER: 2021-2022**
DURATION: 3 Hours **FULL MARKS: 150**

CSE 4303: Data Structures

Programmable calculators are not allowed. Do not write anything on the question paper.
Answer **all** questions. Marks of each question and corresponding CO and PO are written in the right margin.

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1. a) Suppose that you have a *reference* to a node in a *singly linked list* and this node is not the last one in the list. 5+5 CO2 PO2
   i. How could you *remove* the element at this node from the list, while maintaining a proper linked list data structure? Note that this would reduce the number of nodes by 1. The solution should not require looking for this node by starting at the head of the list, rather it must be done in constant time i.e., $O(1)$. (The solution is just a few lines of code).
   ii. How could you insert an element into the list at the position that comes *before* the element at this given node?

   b) Consider the polynomial: 4+4 CO2 PO1
   $$5y^2 - 3y + 2$$
   Represent the polynomial as an '*expression tree*' where the internal nodes can only contain the operators. Utilize the newly generate tree to convert the polynomial as *postfix* expression.

   c) Illustrate the process to maintaining a *Queue* using two *stacks*. 7 CO2 PO3

2. a) Suppose a *Hash function* is defined as '$hash(x) = x \ mod\ 10$'. A set of number is given as $(4371, 1323, 6173, 4199, 4344, 9679, 1989)$. Insert the numbers inside a hash table using the following collision resolution strategies: 5×3 CO1 PO1
   i. Separate chaining
   ii. Quadratic probing
   iii. Double hashing; considering a second hash function $f(i) = i \times hash_2(x)$ and $hash_2(x) = 7 - (x\ mod\ 7)$.

   b) Design an algorithm to implement a *First-In-First-Out Queue* using a *Priority Queue.* 10 CO3 PO3

3. a) Canadian postal codes are of the form '$L_1D_1L_2D_2L_3D_3$', where 'L' is always a letter (A-Z) and 'D' is always a digit (0-9). Suppose you have a company and you wish to keep track of customers who have similar postal codes, so the map might be (postal code, list of customers). Let the letters A-Z be coded with numbers 1 to 26, for example $code(B) = 2$, and let the digits be coded by their numerical values. 3+5 CO3 PO2
   i. For a hash function:
   $$hash(L_1D_1L_2D_2L_3D_3) = \left(\sum_{i=1}^{3} code(L_i) + \sum_{i=1}^{3} code(D_i)\right) mod\ 10$$
   Give an example of a postal code that begins with $H3A$ and collides with $H2B5X7$.
   ii. Give an example of a *hash function* that would never result in a collision. How large would the hash table need to be?

   b) Both *hashing* and *binary search trees* allow to efficiently store and access map entries. What are the advantages/disadvantages of each, which would make you choose one over another? 5 CO3 PO2

   c) Given a set of numbers stored in a *Priority Queue*, design an algorithm to add an amount 'x' to the i^{th} location. Here 'x' is an integer value and 'i' represents the index (starting from 1). Assume that the number with the higher value is considered to have the lower priority. 12 CO2 PO3

4. a) Draw all *binary search trees* of height 2 that can be made from *all* the letters ABCDEF, assuming their natural ordering. 5 CO1 PO2

b) Form a *binary search tree* of strings from the sentence: 3+2 CO1 PO1

“$Form\ a\ binary\ search\ tree\ of\ strings$”

In particular, the tree must be constructed by adding the words in the order they are found in the sentence, namely first add the word “Form” to an empty tree, then add the word “a”, etc. Mention the status of the tree after removing the word “Form”.

c) Two *binary trees* A and B are said to be *isomorphic* if, by swapping the left and right subtrees of some nodes of A, one can obtain a tree identical to B. For example, the following two binary trees are isomorphic. 10+5 CO2 PO3

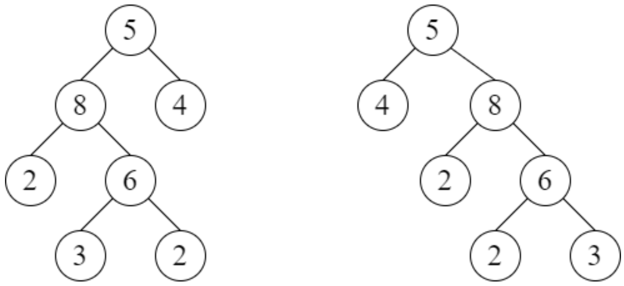

Figure 1: Binary Tree for Question 4(a)

i. Design a recursive algorithm $isIsomorphic(n1, n2)$ for checking if two binary trees are isomorphic. The arguments should start with references to the root nodes of the two trees.

ii. Simulate the proposed algorithm using the trees mentioned in Figure 1 and show that the algorithm is functioning properly.

5. a) *FIFA World Cup 2022* is going on, and as expected, thousands of ‘*pages*’ (with unique names) have been created inside ‘*Facebook*’. Facebook is providing two types of search criteria so that the fans can easily search their desired pages. If the ‘*query*’ directly matches with the title of any existing page, it only shows that in suggestion. Otherwise, it shows all the pages that start with that query string. Discuss an effective approach to satisfy the requirements. (Pseudocodes not required) 7 CO3 PO2

b) $7, 5, 4, 21, 2, 10, 14, 8\ ,1, 9$ 3×6 CO2 PO1

Showing detailed steps, perform the following operations on a *Segment Tree* (Array index starts from 1):

i. Build the tree using the above-mentioned numbers where the internal nodes represent the sum of values within a given range.

ii. Increment all values within the range (4,8) by 5.

iii. Increment all values within the range (1,9) by 10.

iv. Show the summation of the range (2,4).

v. Show the summation of the range (3,7).

vi. Add 7 to the value at index-8.

6. a) Define *Data Structure*. Mention separate use-cases for *eight* different data structures. 2+ 1.5×8 CO1 PO1

b) Consider the following sequence of *stack* commands: 5 CO1 PO1

$push(a), push(b), push(c), pop(\,), push(d), push(e), pop(\,), pop(\,), pop(\,), pop(\,)$

i. What is the order in which the elements are popped?

ii. Considering the positions of the $push(\,)$ commands to be fixed, change the position of the $pop(\,)$ commands in the so that the items are popped in the order: ‘b, d, c, a, e’.

c) How is constant time accessing ensured for an array of integers, whereas, not for Linked Lists? Does this property contradict in case of an array of strings, in which the strings have possibly different lengths? 4+2 CO1 PO2

Name of the Program: B.Sc. in CSE (94) **09 September 2021**
Semester: Winter 2020-2021 **Time: 2:30 PM – 4:00 PM**

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
<u>Department of Computer Science and Engineering (CSE)</u>

SEMESTER Final EXAMINATION **Winter Semester: 2020-2021**
Course Code: CSE 4303 **Full Marks: 75**
Course Title: Data Structures **Time: 1 Hour 30 Minutes**

There are **<u>03 (three)</u>** questions. Answer all of them. Figures in the right margin indicate marks. The examination is an **Online** and **Closed Book.** Marks of each question and corresponding **CO** and **PO** are in the brackets. Renaming format 'fullID_cse4303.pdf'.

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1. Showing detailed steps, perform the following operations for an AVL tree. 10+5+5+10 CO1 PO1

$$(ID+50), 100, 175, 30, floor\left(\sqrt{ID}\right), 50, (ID\%10), 15, (ID^2), birthdate$$

   i. Insert the above-mentioned numbers in the tree. [Mention the $balance\_factor$ with each node in brackets]
   ii. $Successor(ID)$ [simulate properly with justification on how to reach the value]
   iii. $Delete(root)$, Replace with $Tree\_maximum$.
   iv. $Inorder(root)$: Traverse the whole tree. [Simulate by showing the different stages of the recursion stack]

   Note:
   - $'ID'$ means the last two digits of your $student\_id$
   - '$birthdate$' is the 'dd' from your date of birth considering $'dd/mm/yy'$ format.

2. Show detailed steps perform the following operations: 20 CO1, CO2 PO1,PO2

$$(ID+50), 100, 175, 30, floor\left(\sqrt{ID}\right), 50, (ID\%10), 15, (ID^2), birthdate$$

   i. Build a $Segment\ Tree$ using the above-mentioned numbers where internal nodes will represent the Sum of values within a given range.
   ii. Increment all the values within range [2,5] by $'ID'$.
   iii. What is the summation of the range [4,9]?
   iv. Increment all the values within range [6,8] by $'floor\left(\sqrt{ID}\right)'$.
   v. What is the current value of $index-5$?
   vi.

   Note: Array index starts from 1.

3. a) i. Suppose instead of quadratic probing, we use 'cubic probing'; here the i^{th} probe is at $hash(x)+i^3$. Does it improve on quadratic probing? 5+5 CO1, CO4 PO1,PO3
   ii. Explain the concept of '$Rolling\ Hash\ Function$' in the context of the $Rabin\ Karp\ Algorithm$. Simulate its benefit by searching the string '$abcd$' in your $full\_name$. (Show only the first three attempts.)

   b) What are the benefits of using $Trie$ over $Hash\ Tables$ for storing string? 5 CO1, CO2 PO1,PO2

   c) What is Data Structure? Why do we care about it? Write an appropriate use case for each of the data structures taught in this course. 10 CO1, CO3 PO1,PO3

Islamic University of Technology (IUT)
Organisation of Islamic Cooperation (OIC)
Department of Computer Science and Engineering (CSE)
Semester Final Examination (Online)

CSE 4303: Data Structures

Duration: 1 Hour 30 Minutes + 10-15 minutes for submission

Full Marks: 75

Instructions:

- Write your Name, Student-ID and Course Code on the top of the first page. Maintain a serial number on the Top-right corner of each page.
- Answer all the questions. Figures in the right margin indicate marks.
- Sit in proper position and maintain the environment as per the Guidelines.
- No examinee is allowed to scan the file unless 1 hour 30 minutes is finished.
- For any circumstances, follow the instructions of the invigilator.

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1 The following set of numbers represent a Max Heap. 5

A = {15,13,9,5,12,8,7,4,0,6,2,1}

Perform the operation 'Heap_Extract_Max(A)' once.

2 Why is the process of building a Max heap from an arbitrary set of numbers designed in 'Bottom-up' approach? 5

3 Suppose an AVL Tree is used to implement a Min Priority Queue. What will be the complexity of the following operations: 3

a) push(x)
b) pop()
c) minimum()

4 Let T be a Binary Search Tree whose keys are distinct. Let x be a leaf node, and y be its parent. Justify that $y.key$ is either the smallest key in T larger that $x.key$ or the largest key in T smaller than $x.key$. 7

5 Build a Segment Tree using the following set of numbers for finding the Maximum value amongst a given range: 10+5

{10,5,7,-2,8,14,3,0,1,-12}

Show the detailed steps of recursion along with the step by step development of the Segment Tree.
Finally, show the detailed steps for finding the Maximum value for the range (3...7).

6 A Hash function is defined as '**h(x)=x mod 10**' A set of number is given as follows: 5+10

S={4371,1323,6173,4199,4344,9679,1989}

Show the resulting:

a) Separate chaining Hash Table.
b) Hash Table with a second hash function for solving collision:
   $f(i)=i*hash_2(x)$ and $hash_2(x)=7-(x \bmod 7)$.

7 Implement a Dictionary using Trie data structure which will include a set of words. The dictionary allows the following features: 3×5

a) Allows Spell-checking and produces an error message if misspelt.
b) When the prefix of any word is inserted, it shows the number of words that starts with that prefix.

Write pseudocodes of the necessary functions for building the dictionary and implementing Features (a) & (b).

8 Let's assume the Dictionary mentioned in Question-7 is now stored in a Hash-Table. The Hash function is represented by, 10

$$Hash(word) \sum_{i=0}^{wordSize-1} word[wordSize - i - 1] \times 37^i$$

Let's say the Hash table will is represented by an Array for size 100. Each word is mapped at a specific location using the Hash function. However, if two words are mapped to the same index, Linear Probing will be considered as the Collision Resolution method.
Design the algorithm and write necessary pseudocodes to satisfy the requirements.